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Physical AI in Construction

Journey to Unmanned Construction
powered by AI & Robotics

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We Created the First and Only Full-Stack Physical AI Solution in Construction

Market-Proven Solution

2.7x

Growth Strong Vertical Lock-in

- Achieved average **YoY revenue growth (3M$ in '23, 7M$ in '24, 21M$ in '25)**, with strong Lock-in
- Already validated Physical AI solution in conservative construction verticals which is **8.7T$ market.**
- **Number of solutions with a client: x19** (3 ->57) in 20 months

High-Profitability BM

>80%

GM & Break-Even Achieved

- Monthly BEP achieved in '25; on track for full-year BEP in '26
- As the business transitions to a **subscription-based model, profitability is expected to improve even further.**

Full-Stack Physical AI (HW+SW)

104K hrs data

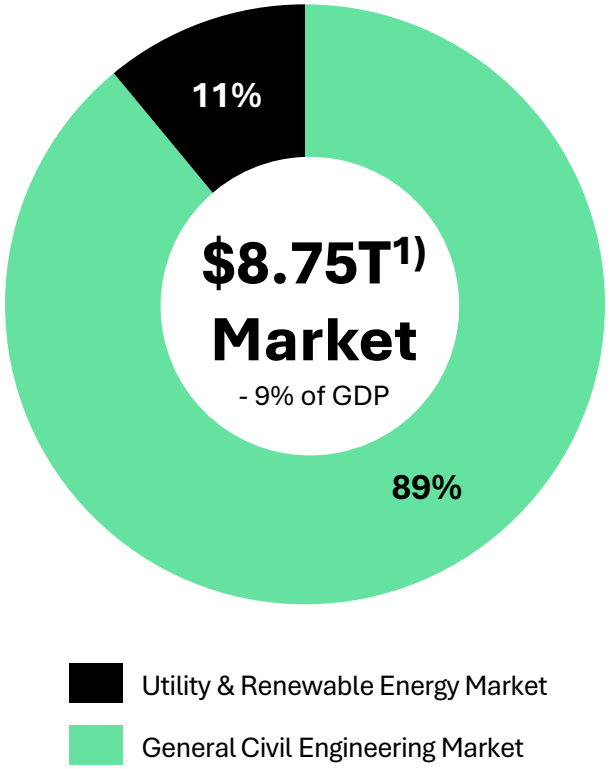
3M+ Hours
Data until 2028

- Unique player with an end-to-end **HW + SW Physical AI stack with rapid vertical expansion.**
- Capturing **3M+ hours of proprietary physical operation data in 3 years**, fueling AI accuracy and creating an unrivaled data moat for unmanned construction.

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A $8.75T Market Facing an Urgent Problem

Massive Market



Massive Pain



- ✓ **Severe Labor Shortage**
  - **440K U.S.** construction jobs unfilled (2025)
  - **41%** workforce retires by 2031- AGC.org
- ✓ **Margin erosion from rising labor and material costs**
  - Construction material costs(PPI) up
  - **40%+** from 2020- AGC.org

1) Civil Engineering market size and share (2025)
(https://www.expertmarketresearch.com/reports/civil-engineering-market)

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Xpanner's Answer: Automation-as-a-Service via Physical AI

Automation-as-a-Service

Core concept

- **Upgrade instantly**
  turn any machine into smart, automated equipment.
- **Expand easily**
  add new tasks anytime with simple updates.
- **Work smarter**
  safer, faster, and fully supported operations via OTA*

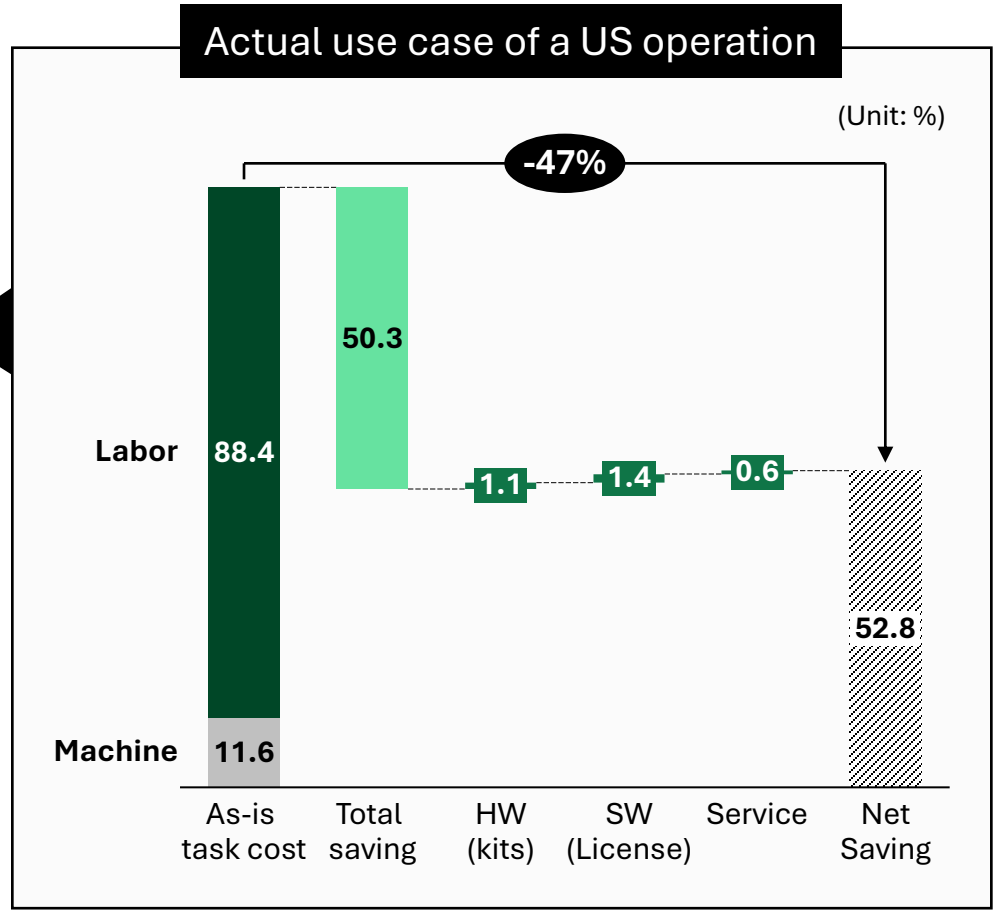
*OTA: Over the air

Proven benefit

47% ↓
Reduction in Total Costs
(Mainly labor)

51% ↓
Shorter Project Duration

Improved On-site
SAFETY ↑



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Since ‘24 in US: Rapid Customer Wins, Fast Revenue Expansion, and Zero Churn

	Customer Tenure	Initial $		**Cumulative $**	Annualized growth	# of project
Client Q	18 months	0.2M$	→	**5.8M$ (29x)**	19.3x	6
Client M	10 months	1.2M$	→	**4.3M$ (3.6x)**	6.2x	4
Client B	9 months	0.6M$	→	**2.8M$ (4.7x)**	9.4x	2
Client D	4 months	1.7M$	→	**1.7M$**	1	1
Client G	4 months	0.7M$	→	**0.7M$**	1	1
Client S	2 months	1.6M$	→	**1.6M$**	1	1
Client B	2 months	1.9M$	→	**1.9M$**	1	1
⋮	⋮	⋮		⋮	⋮	⋮

0%
Churn

18 months in U.S: 62 EPCs in pipeline incl. all Top 10

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Market-Proven with Growing Recurring Revenue

275%

Ave. YoY Revenue (2023-2025)

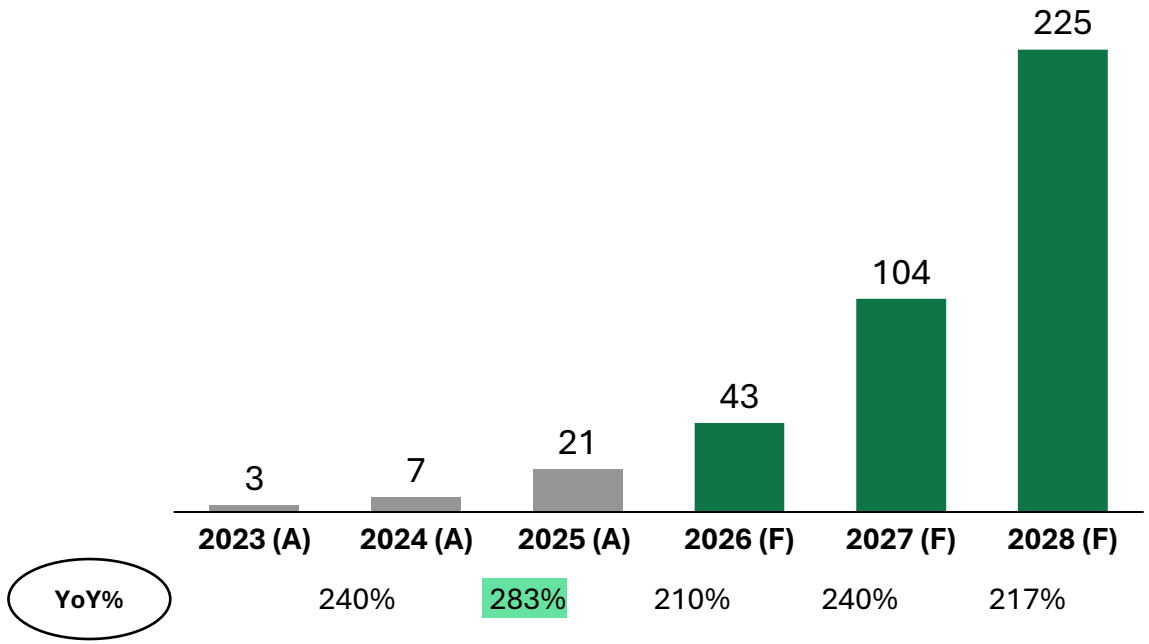
$42M

Cumulative Revenue (~2026. 1Q)

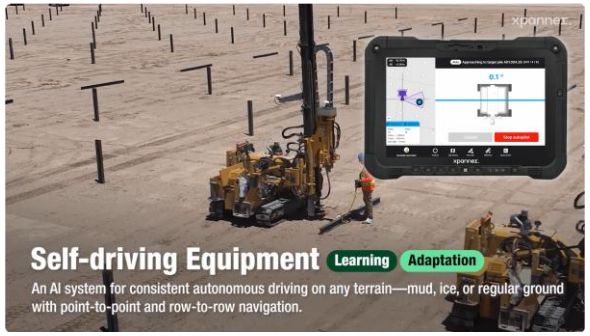
90%

US revenue portion

(Unit: M$)



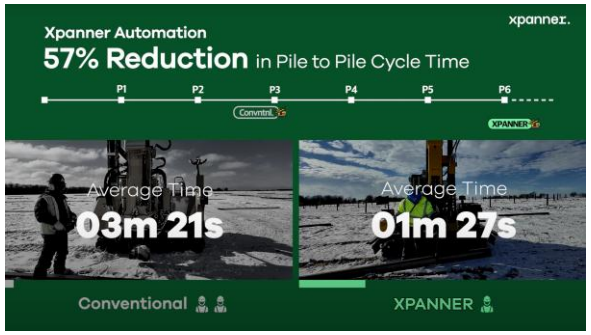
The Full-Stack Solution for Intelligent Construction



X1 Automation Kit
Physical AI Module (H/W)

Our core retrofit kit equips existing machinery, transforming it into a Software Defined Machine (SDM)

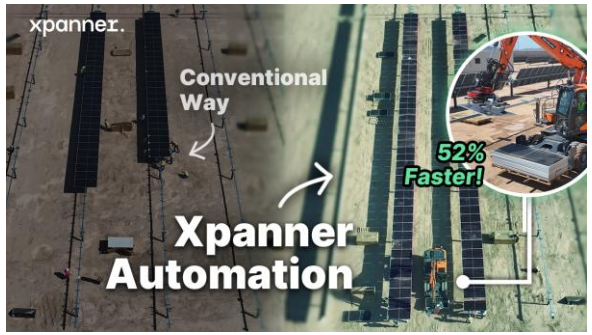
▶ Click to Play Video



X1 Piler
Piling Automation Solution

Our first commercially successful application, proving a 57% reduction in pile-to-pile cycle time in solar construction

▶ Click to Play Video



X1 Automation Kit
Material Handling Solution

Our core retrofit kit equips existing machinery, transforming it into a Software Defined Machine (SDM)

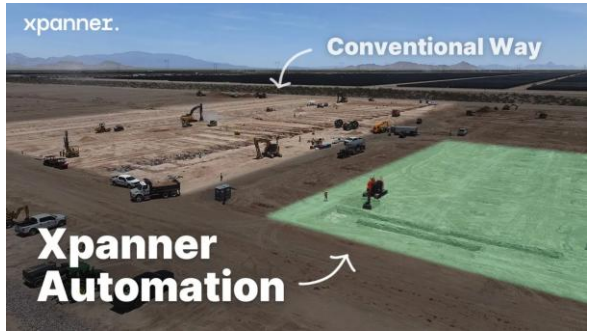
▶ Click to Play Video



Xpanner Connect
Digital Twin Module (S/W)

Collects vast data (progress, accuracy, utilization, etc.) and enables real-time synchronization between project info and machine data.

▶ Click to Play Video



Civil Automation Solution
BESS Solution

Expanding our proven technology to high-growth civil sectors like Battery Energy Storage Systems (BESS) Construction

▶ Click to Play Video

Product Offering

Our Current Offering : Triple-Module AaaS

Physical AI Module

Full-stack physical AI

· HW ·

SDM Retrofit Kit
(Software Defined Machine)



Main processing unit **Mango/M2**



Hyd. System upgrade kits

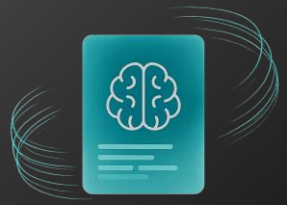


Sensor fusion

+

· SW ·

Automation License



Real-time updating **License**

Task-specific

One machine can have multi licenses

Digital Twin Module

Real-time data collection



- Collect data: Project info, machine date, progress, accuracy, utilization, motion, etc.
- Showing real-time data synchronization of project info. with machines data

Field Service Module

Tech-based service



- Provide clients with essential 'Accuracy Assurance Routine' where Field Ops, QC, and AI teams collaborate daily
- Remote service to improve system continuously (CI/CD concept in physical world) + AI agent

Perpetual ($90K) or Subscription ($10K per month)

$1K per site/month

Up to $25K per site/month

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From a Modular Approach to a Scalable Subscription Engine

- Through business model diversification, we can achieve faster growth and accelerate revenue generation

Revenue Model

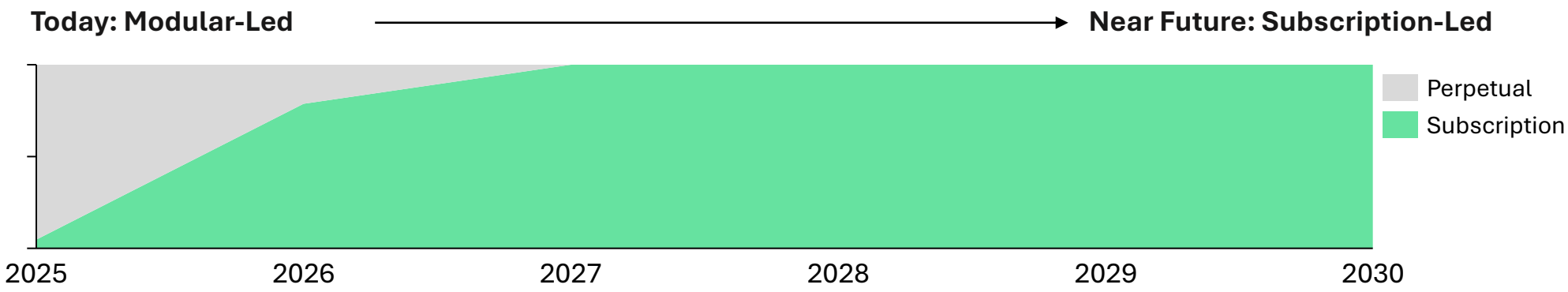
Revenue Model	Perpetual Modular Model		All-in-One Subscription Model
Description	**H/W: X1 Kit**	**S/W: Xpanner Connect**	**Complete Solution Package**
	Core H/W Package	Real-time data sync	No upfront costs, scalable deployment
License Type	Perpetual	Subscription	Subscription
Pricing	$90K (One-Time)	$1K/month	$10K+/month
Initial Investment	***High***		***None***
Scalability	***Incremental***		***Exponential***

Expected Benefits

For Xpanner
- Rapid market penetration and **accelerated profitability**
- Accelerated **physical data acquisition**

For Customers
- **Reduced initial investment burden**

Revenue Mix Evolution



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Entered the market with our Automation Kit Today and Scale into “Physical AI OS”

	Short-term (2025-2026)	**Mid-term** (2027-2028)	**Long-term** (2029-2031)
	Accurate Data Collection while providing AaaS	**Self-enhancing machines with field-feedback-loop**	**Physical AI Platformization**
Physical AI module	• Expansion: Task-specific, equipment • Develop M2 for accurate data collection + enhance recognition • In-house CORE components	• Expansion: By process, per system • Plug & play module gen 2 with M2 • OEM standard: API/Interface • Robotics: IP-grade, multi-industry use	• **Physical AI OS** connecting equipment, robots, sensors, S/W, and AI: open for all Mfg./Software companies
Digital Twin module	• Real-time progress/predict/ analysis based on data(X1 kits + API): Data snowball • Cloud-based automation license model	• Predictive/Preventive information service based on real-time data based on AI model • Remote operation feature launching	• **Digital Twin** for pre-con simulation, training physical AI, and real-time unmanned operation
Field Service	• Pre-con + Construction service • Field feedback loop to ensure ‘Accuracy Assurance Routine’ • All communication log will be used for training AI agent	• Expand service channel network based on “AI agent + wearable” • Sensors and machines will replace some of human's report	**→ Achieve several ‘Full Unmanned Processes’**

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Tech Vision

"Productization of Automation" — *"Physical AI based Automation"* — *"Physical AI based unmanned"*

2025 | 2026 | 2027 | 2028 | 2029 | 2030

Roadmap

1 Foundation

Goal: Build the base for task-specific automation.
Result: Reliable retrofit control platform powering **X1 kits for solar projects.**

3 Integration / Expansion

Goal: Link every machine to the cloud.
Result: Cloud-connected, data-driven construction ecosystem.

2 Enhance CORE

Goal: Turn the controller into a full industrial HW/SW platform.
Result: Unified platform ready for AI-based perception and control.

4 Physical AI OS (Autonomy)

Goal: Achieve full autonomous operation.
Result: Physical AI OS unmanned, self-operating construction sites.

1
- Developed **Mango**, the first-generation controller for pile-driving automation.
- Integrated **GNSS + IMU** for accurate positioning and motion control.
- Collected real-world operation data to train future AI models.

2
- Added **AI computation**, **multi-sensor fusion**, and **vehicle networking**.
- Built **QNX RTOS + ROS middleware** for real-time control.
- Enabled **Docker/WASM** container deployment and **quantum-secure communication**.

3
- Released **Xpanner Connect** with web HMI, remote control, and 3D visualization.
- Enabled real-time task assignment and productivity tracking.

4
- Deploy **task-specific AI modules** across machine types.
- Enable **cloud–machine collaboration** and **multi-agent coordination**.

Core Technical Focus

Develop M2 HW/SW Platform real-time sensor fusion, precision control	**Integrate AI-driven control** into ROS/RTOS loop, route planning, safety management	Enable AI decision-making, multi-agent coordination, Digital Twin-based site control

Representation System

X2 Retrofit Kit, Mangosteen Controller v1	**Xpanner Connect, Automation Apps**	**Physical AI M2 Platform, Unmanned Construction OS**

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Tech Stack for an Unmanned Site by Physical AI

X3 – PHYSICAL AI INFRASTRUCTURE
(2028 – 2030+)

- AI-driven perception & prediction, deterministic planning & control
- Multi-machine coordination with cloud–edge collaborative control architecture
- Unmanned, self-operating construction ecosystem for specific construction tasks

XPANNER CONNECT
(2024 – 2027)

- Cloud-based project management platform for site planning and progress tracking
- Connected fleet data pipeline for continuous learning and OTA model updates
- WebRTC-based tele-operation + WASM-based 3D GIS visualization engine

X2 – MANGOSTEEN PLATFORM
(2025 – 2027)

- Ruggedized all-in-one SDV platform with perception, positioning, and connectivity
- QNX + ROS2 + Custom Middleware with cuPQC-based quantum-safe layer
- Strategic and efficient data collection for model training

X1 – MANGO PLATFORM
(2023 – 2024)

- Distributed architecture of I/O, positioning, machine network, and connectivity
- RTOS-based controller for electrification and task-specific automation retrofitting
- Field-tested data for AI development and early automation (2025 onward)

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Automation Opportunities: 1) PV Plant

Xpanner has a solution but not scope of current phase | Focus area for collaboration | ☑ In-place

Automation opportunities in PV plant

- Site Preparation & Civil works
- Foundation & Pile Driving
- Mounting Structure Installation(Reck)
- Solar Pannel Installation
- Electrical System Installation
- Substation & Grid Connection
- Testing & Commissioning
- Operation and Maintenance

Identified automation opportunities	Xpanner's technology candidates	Product Launch
☑ Topo Surveying with Drone & Vehicle	• Drone solution integrated to Xpanner platform • X1* Kit Gen1 + Utility vehicle	• 4Q '25 • Ready
☑ **Pile Placing (Shakeout)**	• X1 Kit Gen1 + Compact Track Loader • Unmanned dedicated machine	• Ready • 4Q '25
☑ **Pile Driving**	• Orteco machine • X1 Kit Gen1 + Pile Driver	• On sale • On sale
☑ Trenching and Cabling	• X1 Kit Gen1 + Excavator with attachments	• Ready
☑ **Pannel Lifting**	• X1 Kit Gen2 + Excavator with attachments	• 1Q '26
Pannel Lifting & Mounting	• X2** Kit Gen1 + Excavator + Robot arm + sensor fusion	• TBD (~3Q '27)
☑ Overall Site Management Platform	• Cloud-based data platform • API Integration or Xpanner IoT (eg. Procore)	• On sale • 2Q '26

* X1 : Mango Based
** X2 : M2 Based

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Automation Opportunities: 2) BESS Construction

Xpanner has a solution but not scope of current phase | Focus area for collaboration | ☑ In-place

Automation opportunities in BESS Construction

- Site Preparation & Civil works
- Grounding & Cable Trench Excavation
- Conduit Installation & Final Grading
- Battery Container / Rack Installation
- Electrical System Installation
- Substation & Grid Connection
- Testing & Commissioning
- Operation and Maintenance

Identified automation opportunities	Xpanner's technology candidates	Product Launch
☑ Topo Surveying with Drone & Vehicle	• Drone solution integrated to Xpanner platform • X1* Kit Gen1 + Utility vehicle	• 4Q '25 • Ready
Grading	• X1 Kit Gen2 + Dozer + Grading	• 3Q '26
☑ **Trenching and Grading**	• X1 Kit Gen1 + Excavator + Trenching • X1 Kit Gen2 + Excavator + Trenching	• On sale • 2Q '26
☑ **Backfilling and Compacting**	• X1 Kit Gen1 + Excavator +Backfil/Compact • X1 Kit Gen2 + Excavator + Backfil/Compact	• On sale • 2Q '26
☑ **Pile Driving for Battery Container Foundation**	• X1 Kit Gen1 + Pile Driver + Low Piling	• On sale
Battery Container Lifting & Placement	• X2** Kit Gen1 + Crawler Crane + Heavy Lifting	• 2Q '27
☑ Overall Site Management Platform	• Cloud-based data platform • API Integration or Xpanner IoT (eg. Procore)	• On sale • 2Q '26

* X1 : Mango Based
** X2 : M2 Based

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Automation Opportunities: 3) Data Center

Xpanner has a solution but not scope of current phase | Focus area for collaboration | ☑ In-place

Automation opportunities in Data Center

	Identified automation opportunities	Xpanner's technology candidates	Product Launch
Site preparation & Civil works	☑ Topo surveying with Drone & Vehicle	• Drone solution integrated to Xpanner platform • X1* Kit Gen1 + Utility vehicle	• 4Q '25 • Ready
Structural Frame Erection	**Material Staging/Logistics**	• X2** Kit Gen1 + Skid Steer Loader + Staging	• 4Q '26
Building Envelope	**Material Handling**	• X1 Kit Gen2 + Excavator + Material Handling	• 2Q '26
Interior Finishing	Cabling & Installing	• X2 Kit Gen1 + Robot + Sensor fusion (TBD)	• TBD (3Q '27)
MEP (Mechanical, Electrical, Plumbing)	Material Lifting & Placement	• X1 Kit Gen2+ Boom Lift + Material Lifting	• 4Q '26
Equipment Installation	**Rack Staging/Logistics**	• X2 Kit Gen1 + Skid Steer Loader + Staging	• 4Q '26
Rack & IT Infrastructure Install	Overall site management platform	• Cloud-based data platform which can integrate any type of system with API or Xpanner IoT (eg.Procore)	• 2Q '26
Commissioning & Operation			

* X1 : Mango Based
** X2 : M2 Based

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Automation Opportunities: 4) Power Distribution

Xpanner has a solution but not scope of current phase | Focus area for collaboration | ☑ In-place

Automation opportunities in Power D

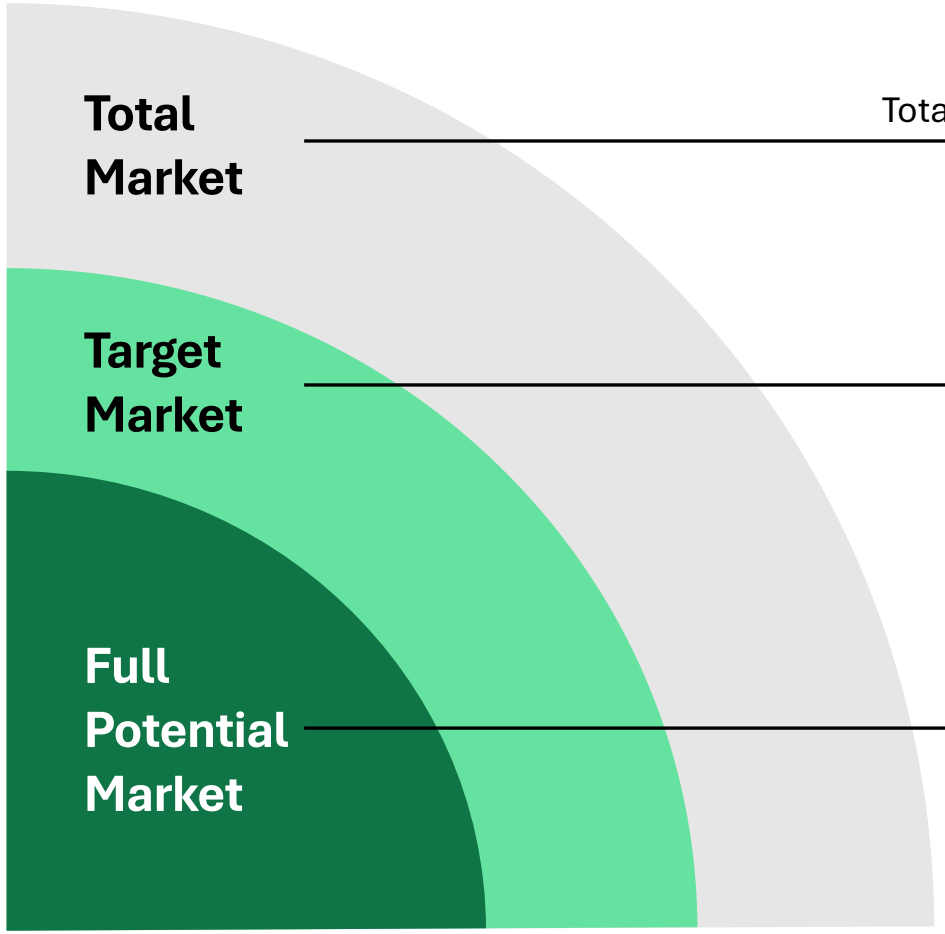
- Site preparation & Civil works
- Foundation & Equipment Pad Construction
- Underground Cabling & Duct Bank Installation
- Overhead Line & Pole Installation
- Equipment Installation
- Electrical & Commissioning
- System Integration & Connectivity
- Operation & Maintenance

Identified automation opportunities	Xpanner's technology candidates	Product Launch
☑ Topo surveying with Drone & Vehicle	• Drone solution integrated to Xpanner platform • X1* Kit Gen1 + Utility vehicle	• 4Q '25 • Ready
Heavy Duty Pile Driving	• X2** Kit Gen1 + Rig Piier + Piling	• 1Q '27
☑ **Trenching and Grading**	• X1 Kit Gen1 + Excavator + Trenching • X1 Kit Gen2 + Excavator + Trenching	• On sale • 2Q '26
☑ **Backfilling and Compacting**	• X1 Kit Gen1 + Excavator +Backfil/Compact • X1 Kit Gen2 + Excavator + Backfil/Compact	• On sale • 2Q '26
Erecting, aligning and installing poles	• X2 Kit Gen1 + Crane + Heavy Lifting	• 2Q '27
Heavy Equipment handling & mounting	• X2 Kit Gen1 + Crane + Heavy Lifting	• 2Q '27
Overall site management platform	• Cloud-based data platform which can integrate any type of system with API or Xpanner IoT (eg.Procore)	• 2Q '26

* X1 : Mango Based
** X2 : M2 Based

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Xpanner Provides AI-driven Construction Sectors First : AI/Data Center/Energy related sectors



Total Addressable Market (TAM)

Total Construction/Infrastructure Market

$3,687B

Current Target Market

AI/Data Center/Energy Related Sectors

$1,025B

Addressable Market

Average 47.6% of Total cost (Labor + Equipment)

$488B

Description: We define the full potential market as the total annual expenditure on labor and equipment within our target segments. This represents the core operational costs that our solution directly impacts and optimizes.

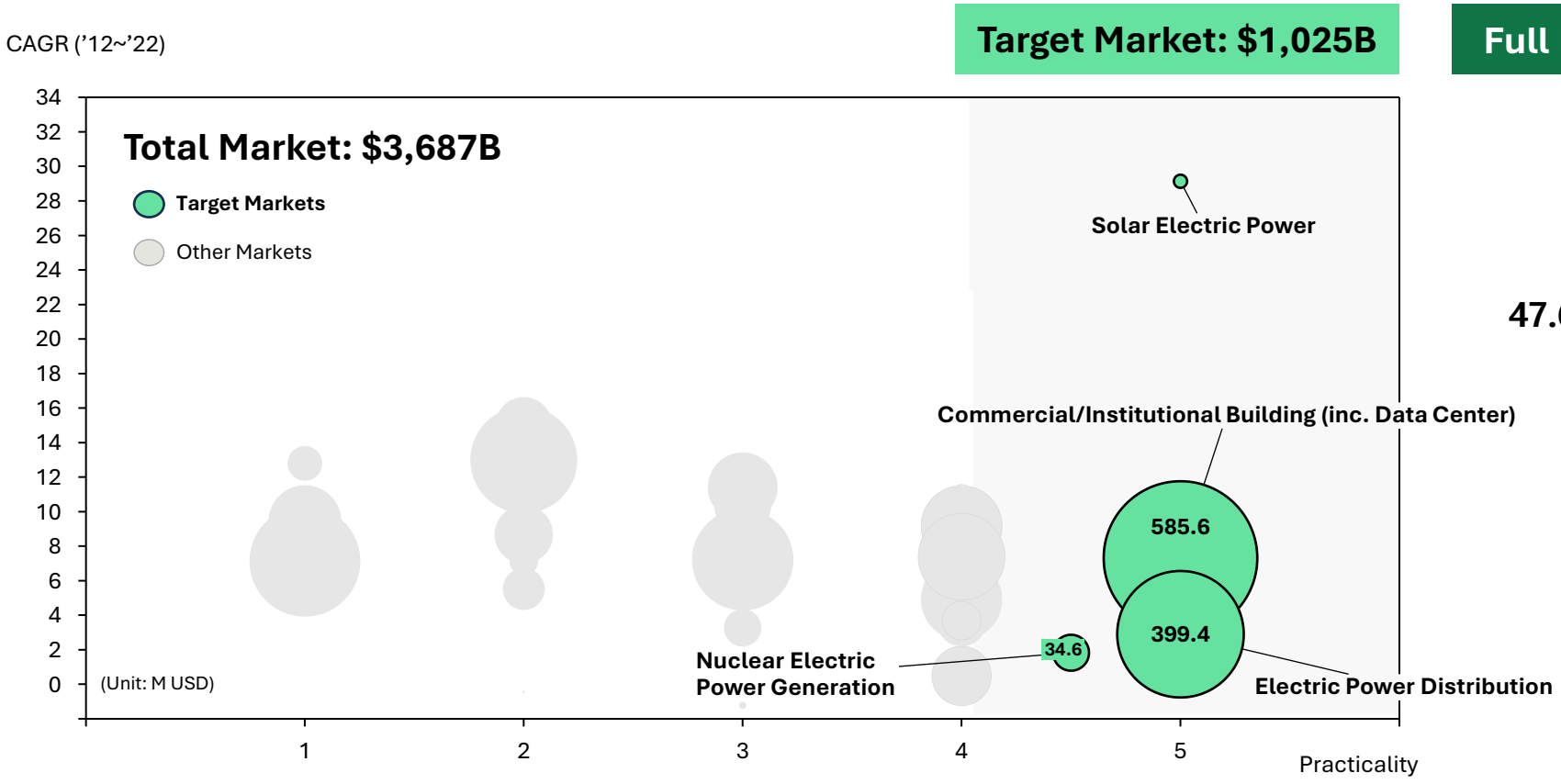
Source: Census – Annual Business Survey (ABS), Economic Census

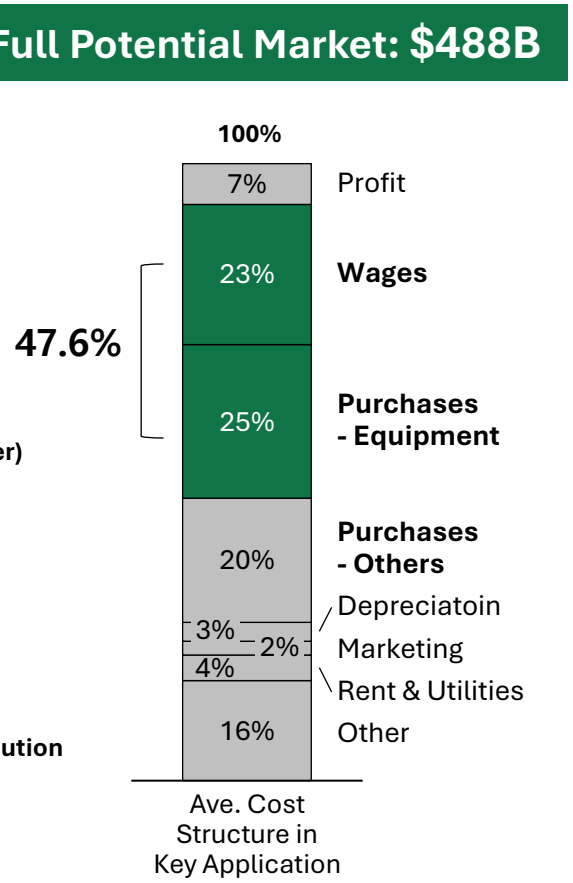
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[Back-up] Strategic Market Prioritization Matrix

Xpanner provides **AI-theme driven construction sectors** first: **AI/Data Center/Energy related sectors**

Strategic Market Prioritization Matrix – Growth / Practicality





Census – Annual Business Survey (ABS), Economic Census
* Source: IBISWorld report 23799 Heavy Engineering Construction in US

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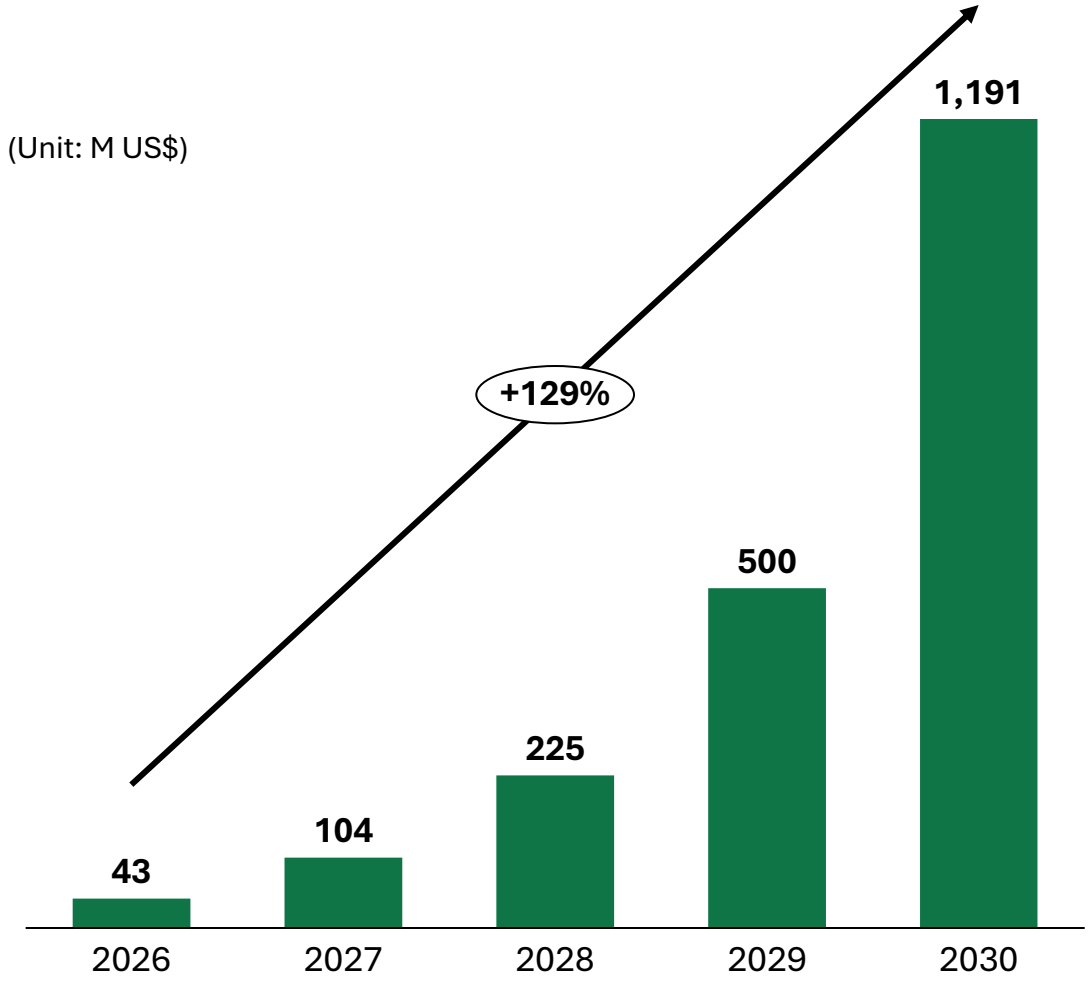
5-Axis Strategy: Lock-In ▶ Enhance ▶ Dominate

Axis	Short-term ('25~'26): Lock-in	Mid-term ('27~'28): Enhance	Long-term ('29~'31): Dominate
Marketing	• **Early EPC wins** in US/KOR • Standards PoC & **strong references** from high-profile case study • Scale-up with **Market lock-in**	• Global EPC/OEM alliances • Regional expansion • Verticals (renewables, data centers)	• Market leader approach • Influence policy/standards • **Global adoption of Xpanner OS**
Technology	• Develop M2(Mangosteen V1 platform) • Cloud license • **Data snowball(pipelines in all aspect)** • **Start in-house production (Core items)**	• Task-specific AI modules • API/interface std. • 70–80% self-production • **IP-grade robot development**	• **Physical AI OS** • Full unmanned process capability • 3rd-party integration
Product	• X1 kits • **Equipment expansion (Pile → Excavator → Compact Loader → Dozer)**	• Multi-equipment automation kits • Predictive maintenance • Digital twin services • **Robotics entry service**	• **Construction App Store** • Expand unmanned equipment portfolio
Organization	• Agile Field Ops + feedback loop • Strengthen sales & deployment • Small-scale mfg. line	• R&D hubs in key markets • Strategic alliances • Scale manufacturing capacity	• Global partner ecosystem • Certification programs • Industry training standards
Customer Value	• **Automation based "Accuracy Routine"** • KPI dashboards • Subscription (MRO, updates, support)	• **Building block approach for end-to-end style of process automation** • Predictive insights • Global service coverage	• **Integrated unmanned site mgmt.** • **AI insights marketplace** • **End-to-end Physical AI**

Exploring short-term inorganic opportunities to make it faster

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Financial Projection



Key assumption

- **Revenue assumptions:** 1) Units sold × ASP* + 2)Subscription ARPU** × customer base
- Expansion of regional reach and product offerings are considered.

*ASP: Average Selling Price
**Annual Revenue per customer

Visionary Leadership Team with Proven Expertise Driving Sustainable, Tech-Enabled Growth



Henri Lee
CEO & Co-founder
Bobcat

- 14 years at Doosan & Bobcat, leading strategy and new ventures across global markets, including the U.S. and Europe.



David Shin
CTO & Co-founder
VOLVO Construction Equipment

- 15 years at Volvo CE, led automation & digitization, award-winning engineer with deep product development expertise.



Ryan Park
CFO/CSO & Co-founder
Shinhan Bank

- 10 years at Doosan in corporate finance & M&A, plus 6 years as VC director at Shinhan, leading global investments.

Global team's expertise

VOLVO Construction Equipment
Bobcat
HD HYUNDAI INFRACORE
HEXAGON
DASSAULT SYSTEMES
Sony Interactive Entertainment
BOEING
BMW
Shinhan Bank
GE
KPMG

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Xpanner: The Only End-to-End Automation Platform in Construction

	xpanner.	Geospatial Tech companies*	Rental Companies	Equipment Manufacturers	Manufacturer Dealer	Construction Robotics
System & Platform						
Site level system integration	O	O	X	X	X	X
Site Information system(w/ BIM)	O	O	X	X	X	O
Brand extensibility	O	X	O	X	O	O
AI & Automation Technology						
Individual machine motion automation	O	O	X	O	O	O
Fleet automation technology	O	△	X	X	X	X
Learning/Adaption of automation	O	X	X	X	X	O
Go-to-Market & Customer Success						
Forward deployed engineers in service	O	O	O	O	X	O
Task automation features in the market	O	O	O	△	X	X
Understanding construction equipment system	O	X	O	O	O	O

*Machine guidance based companies such as Trimble, Hexagon, Topcon, etc.

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Xpanner: The Only End-to-End Automation Platform in Construction

Category	Xpanner (Physical AI Platform)	OEMs (Caterpillar, Volvo, Doosan)	Built Robotics	Geo-Spatial Tech (Trimble, Topcon, Leica)	Cosmic Robotics	Bedrock Robotics
Core Focus	Full-stack Automation (automation retrofit + RobotOS + data loop)	Factory-built automation & machine sales	Excavator autonomy for earthworks	Positioning & machine-guidance software	AI construction robot arms / manipulators	Specialized mining and underwater autonomy
Business Model	**Automation-as-a-Service** (HW + SW license + data subscription)	Hardware sales + service contracts	Hardware retrofit + rental	Hardware + software license	Prototype robot sales / integration	Niche robot system sales
Hardware Ownership	**In-house controllers (Mango/M2) + sensor fusion modules**	**OEM control units only**	**Retrofit kit based on OEM hardware**	**Third-party sensor platform**	**Proprietary robot arms & actuators**	**Proprietary autonomous vehicles**
Software Stack	**RTOS + ROS middleware + AI control + Digital Twin**	Closed OEM firmware + limited autonomy	ROS-based automation (earthwork-specific)	GNSS/IMU positioning + 2D/3D grading algorithms	Custom motion planning / ML modules	Embedded AI navigation
Hardware–Software Integration	**Designed together - no existing HW can fully enable our SW**	**Partial; hardware-first**	**Limited to specific machines**	**Software-layer only**	**Tight for proprietary robots**	**Tight for own robots, not scalable**
Autonomy Level	**Task-specific → Multi-machine → Full site autonomy**	Assistance / semi-auto	Single-machine earthwork	Operator-assist only	Fixed-task robots	Domain-specific autonomy
Data & AI Feedback Loop	**Continuous field data (22K → 3M hrs) → model updates**	**Minimal / closed**	**Localized training data only**	**No AI feedback loop**	**Lab-trained models**	**Sensor data not commercialized**
Retrofit Scalability	**Any machine, any OEM**	New machines only	Excavators only	Software upgrade only	Low scalability (bespoke robot form factor)	Low scalability (industry specific)
Market Focus	**Renewables · BESS · Data Centers · Civil Infra**	Heavy equipment / new sales	Earthworks / civil construction	Survey & grading support	Modular robotics R&D	Mining / sub-sea operations
Competitive Edge	**Only player combining proprietary SDM HW, Cloud license SW, and data platform for retrofit autonomy.**	Brand trust + scale	Early excavator use-case focus	Precision positioning legacy	Agile R&D	Niche automation expertise

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We Put Capital to Work to Turn Commitments into Reality

~2023 Pre-Series A

$8.6M
Build CORE tech

- Build CORE team for HW/SW
- Selling individual product package with automation features
- **Built CORE technology sets for scale-up (Modularized HW/SW capacity)**

2024 Series A

10.7M
Global Product Market Fit

- US market expansion
- Product launches in renewable sector
- Launched the physical AI module (X1 Kit + Xpanner connect) as a solution-platform

Now: Series B

~ $40M
Unmanned Construction

- Accelerating & Cementing Market Leadership of Physical AI in construction

* F/X (USD/KRW): 1,400

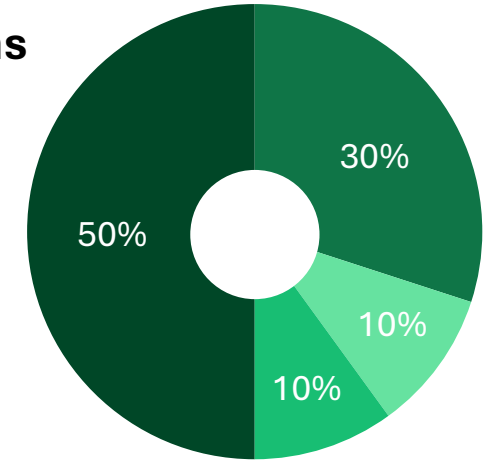
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Strategic Allocation to Solidify Market Dominance

Raising **~$100M**

Acceleration of Unmanned Operations

- **Inorganic:** By building CORE components in-house, **we expedite development while embedding critical engineering know-how** into our M2(Physical AI unit) and sensor fusion capabilities.
- **Organic:** Establishing an engineering facility to develop and test, **reinforcing the Physical AI feedback loop and accelerating development.**



Kickstarting a Virtuous Cycle

- Capital to power **aggressive one-month trial** marketing for our full-stack Physical AI product and to scale a rental + **subscription program into massive ARR.**

Rapidly expand products and markets

- Rapidly scale product offerings and geographic reach

Corporate Growth & Working Capital

Strategic reserves for operational flexibility and risk mitigation

The Impact: From Proven Leader to Industry Standard

1. Speed, know-how, and stronger Physical AI.
2. Capital to fuel trials and scale ARR
3. Build an Unbeatable Physical Data Asset
4. Strengthen key functions to lead key global markets

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Strategic Allocation to Solidify Market Dominance

Category	Allocation (%)	Primary Use	Expected Outcome
1. Product & Platform R&D	**15 ~ 20 %**	• Complete M2 / Mangosteen HW-SW Platform (QNX RTOS + ROS middleware + sensor fusion) • Develop Physical AI OS prototype	• Multi-equipment autonomy; launch of X2 Retrofit Kit (2026) • Comprehensive data collection
2. CORE component Manufacturing	**20 ~ 25 %**	• Localize production of Mango controllers, GNSS/IMU modules • Secure Mfg. engineering and capacity	• Seamless HW–SW integration, enabling next-generation automation performance
3. Data & AI Infrastructure (Xpanner Connect)	**5 ~10 %**	• Build cloud-based Digital Twin / AI training pipeline; expand data from 22K → 3 M hours by 2028	• Higher AI accuracy & recurring data revenue
4. Go-to-Market & Field Operations	**30 ~ 35 %**	• Build a fleet for subscription model-based (Kits and machines) • Expand U.S. Field Ops Team (10 → 40); launch Xpanner Certified Partner Program with EPCs • Establish U.S. PoC hub in Busan & TX	• Deploy 100 + sites; drive ARR to >$50 M by 2027
5. Corporate Growth & Working Capital	**10~15 %**	• Key hires (AI engineers, system integration leads, finance team); operating capital & IPO preparation	• Global scale-up

Technology & Product

- ***Acceleration of Unmanned Operations***

Business model transform

- ***Kickstarting a Virtuous Cycle***

Corporate expansion

- ***Key corporate build activities and reserves***
- ***Geo-graphical expansion***

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Positioned for a Landmark Exit

1st Priority: U.S. IPO

Our goal is a high-multiple IPO on NASDAQ, leveraging the market's strong appetite for high-growth, profitable Physical-AI leaders.

AI Positioning = Valuation Premium

20~44x

AI PSR

3.0x

Traditional Tech PSR

Source: Yahoo Finance, Industry Reports (2025)

2nd Priority: Global M&A

Major industry players are actively acquiring automation tech to complete their platforms, making Xpanner a prime target.

Recent Strategic Moves

- **Black & Veatch** (EPC) acquired **Atonix Digital** (Dec 2023)
- **Komatsu** (OEM) acquired **Moovila** (Q4 2024)
- **Trimble** (Tech) invested in **Civ Robotics** (Feb 2025)
- **Caterpillar** (OEM) acquired **Teleo** (Mar 2025)

E.O.D

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Appendix

Journey to Unmanned Construction
powered by AI & Robotics

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Delivering Value Across the Entire Construction Value Chain with Strong Lock-in Impact

Core Stakeholders: Direct users who gain tangible value and rely heavily on Xpanner's solutions for productivity, safety, and accuracy.

Strategic Integrators: Key players who leverage Xpanner to enhance project visibility and operational efficiency, driving repeat engagement.

Visibility Beneficiaries: Upper-tier stakeholders who benefit from project transparency and data insights enabled by Xpanner's ecosystem.

Xpanner Value Proposition Matrix

Construction Value Chain	X1 Kit (Physical AI)	Xpanner Connect (Digital Twin)	XFO (Service)
PF Financing Capital	• Time saving • Better accuracy	• Ensure Full Visibility into Project and Progress Status	
Developer	• Time saving • Better accuracy	• Ensure Full Visibility into Project and Progress Status	
General Contractor	• Time saving • Better accuracy • Safety	• Ensure Full Visibility into Project and Progress Status	
Sub-Contractor	• Optimized Fleet Allocation • Labor Cost Reduction	• Ensure Full Visibility into Project and Progress Status	• Service Cost Saving
Rental Company	• Enhance Rental Competitiveness	• Ensure Visibility in Fleet Management	• Service Cost Saving
Jobsite Labors	• Faster and More Convenient Operations / Streamlined Workflow		• Faster and More Convenient Operations / Streamlined Workflow

Xpanner Offering

"With Xpanner, we can **monitor our project progress in real-time**. This capability is a game-changer for staying on schedule and ensuring the accuracy."

- VP of Construction Project Management, Customer Q -

"Utilizing Xpanner's solution **increases speed, efficiency and most importantly it boosts the safety** of our crews."

- VP of Innovation, Customer B -

""It definitely **has helped me a lot with surveying,** because I don't have to do as many points. It's a **big help on site**."

- Surveyor, Customer N -

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Since ‘24 in US: Rapid Customer Wins, Fast Revenue Expansion, and Zero Churn

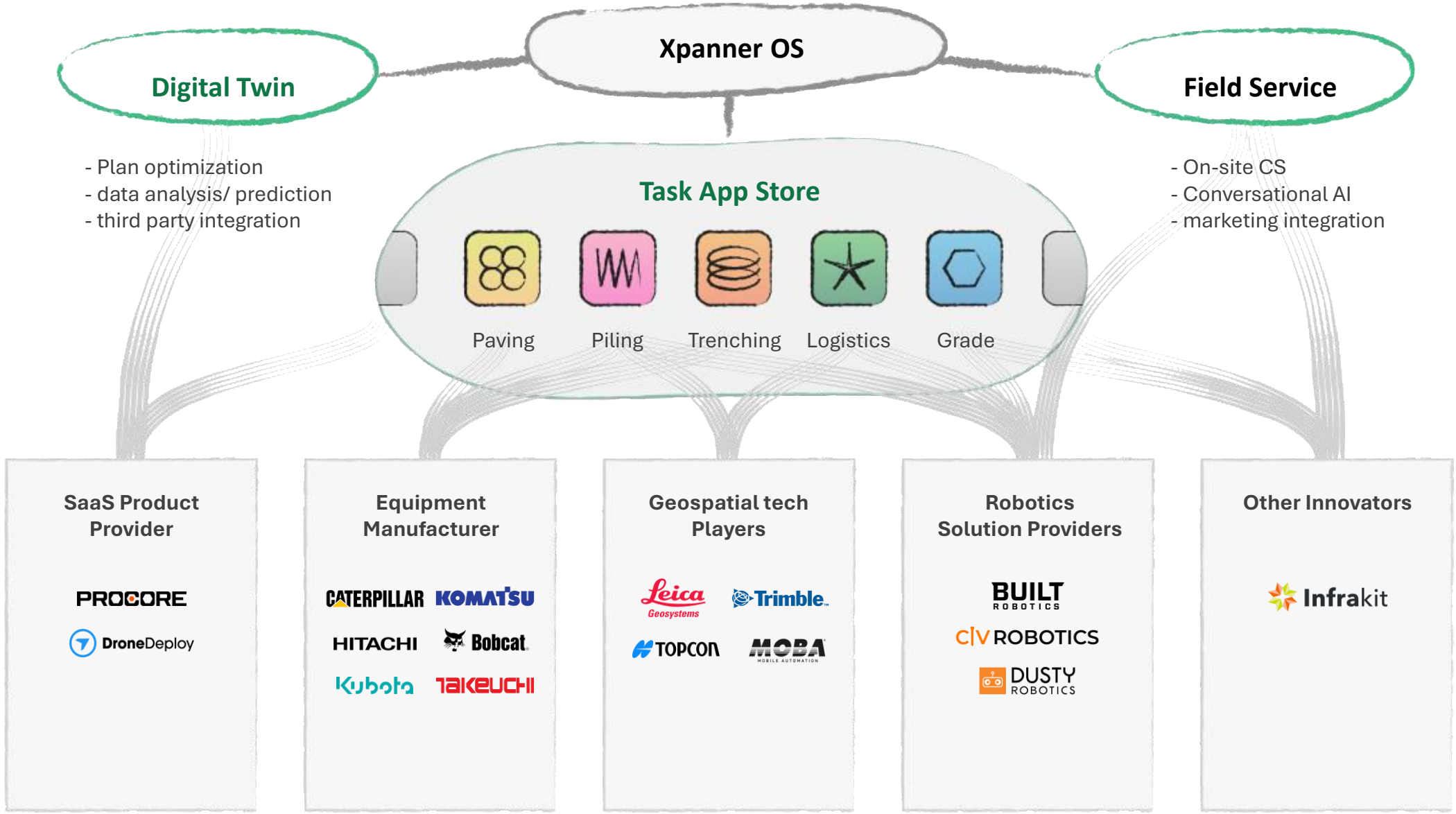
	BB PV	OR PV	AT8 BESS	AT5 PV	AT8 PV('26 1Q)
Site Prep	x	x	Trenching	Trenching	Grading /Trenching
Structure	Piling	Piling	Back filling	Piling	Shake-out /Piling
Installation	x	x	x	Panel lifting	Panel lifting
Electric	x	x	x	x	x
⋮	⋮	⋮	⋮	⋮	⋮

100%
Retention to next project

Expanding to other tasks

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Barrier-Free by Xpanner OS

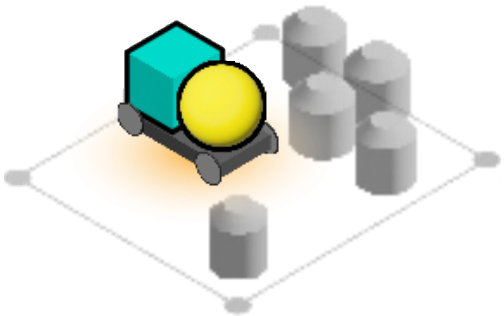


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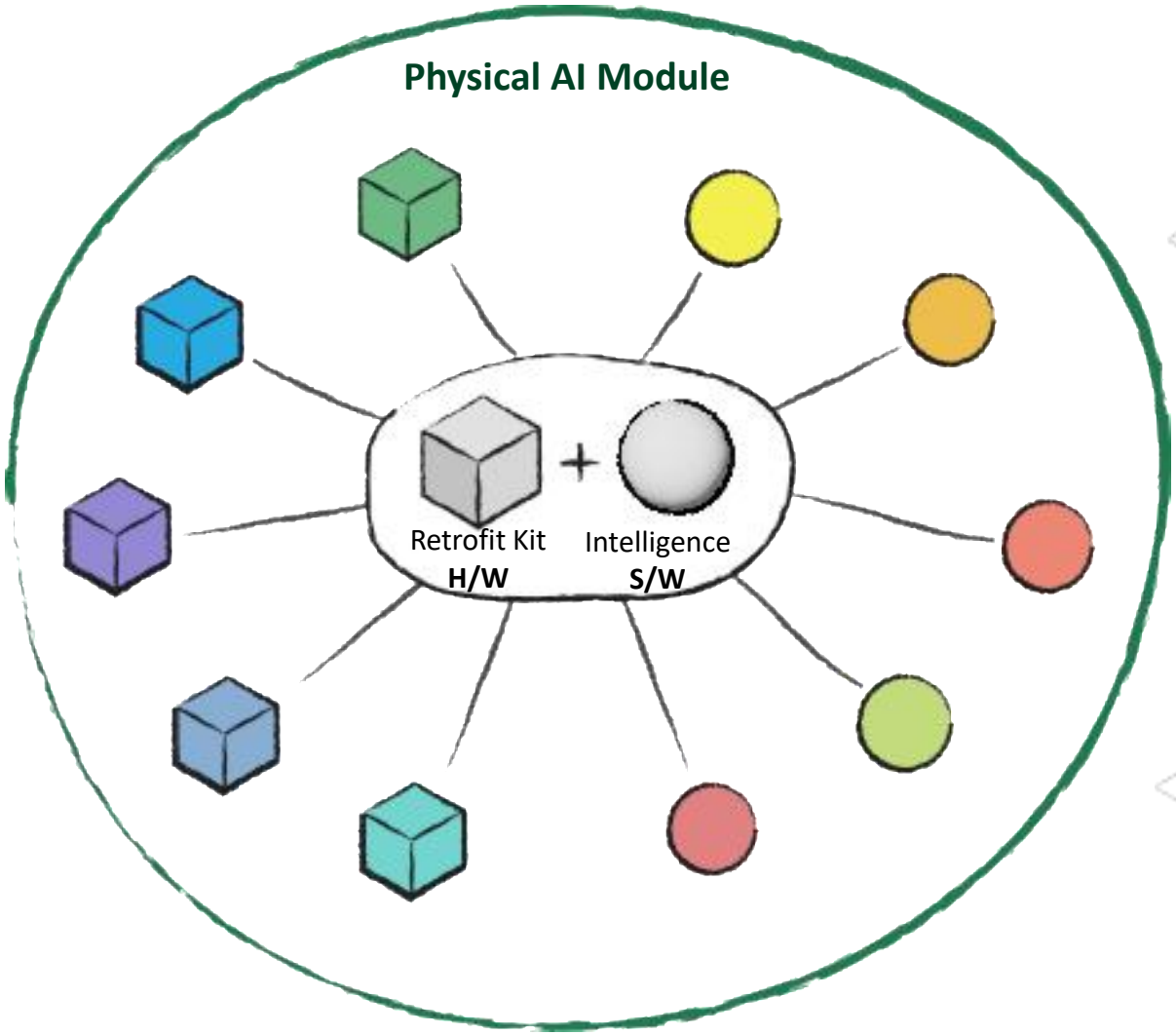
Full-Stack Physical AI – Scalable Modular Approach

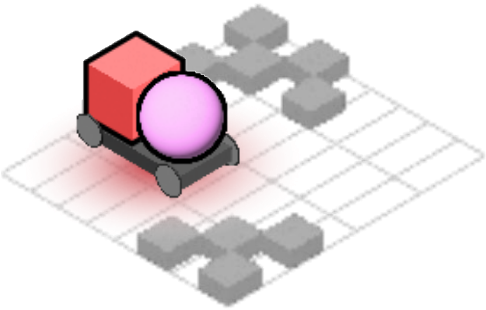


Task A

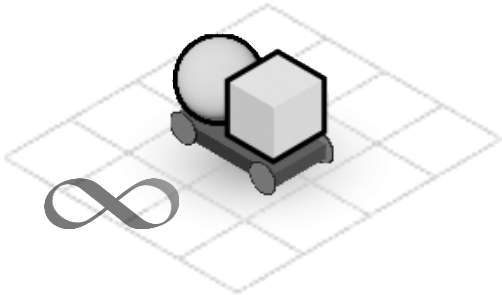


Task B





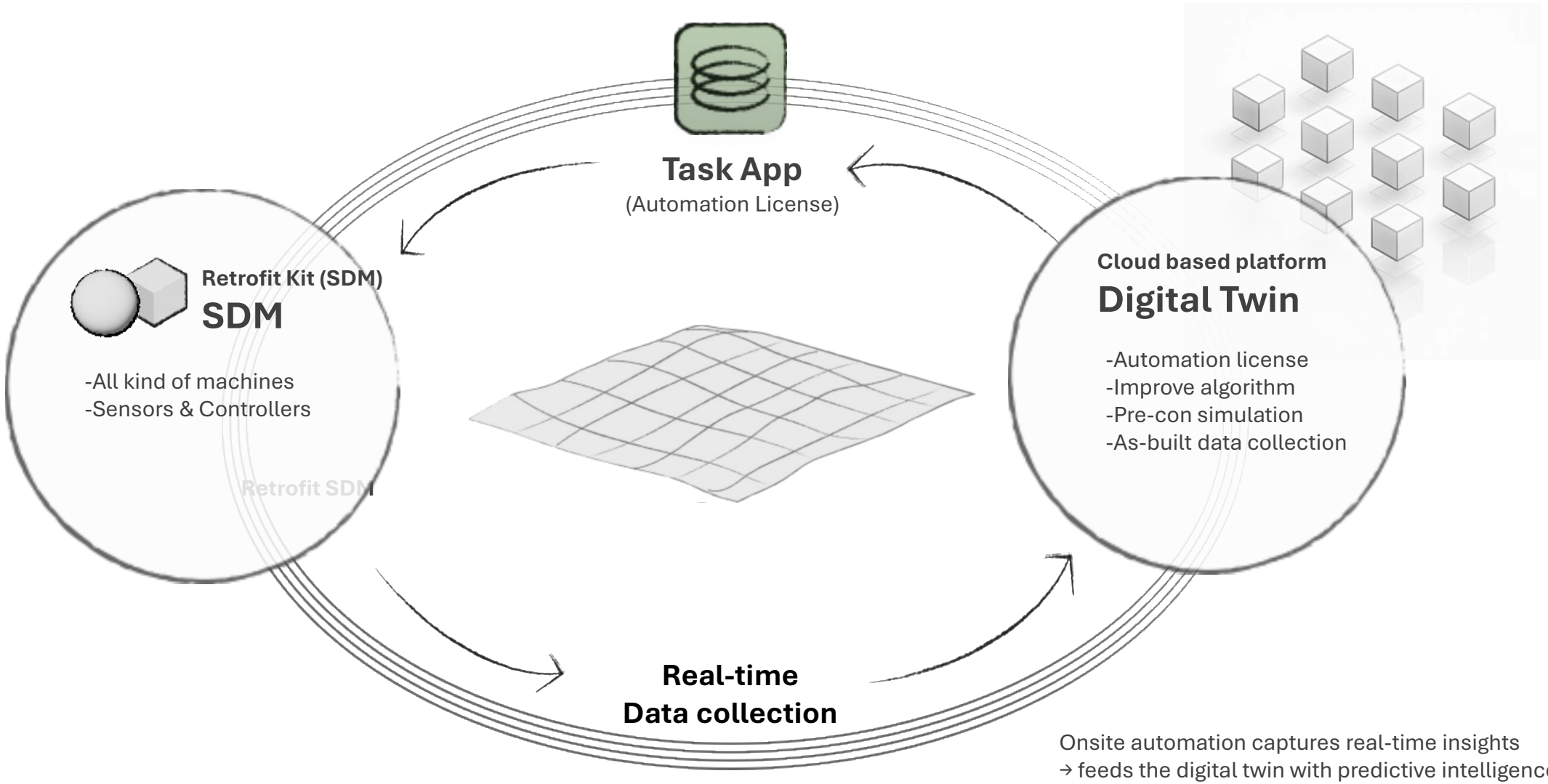
Task C



Infinite ++

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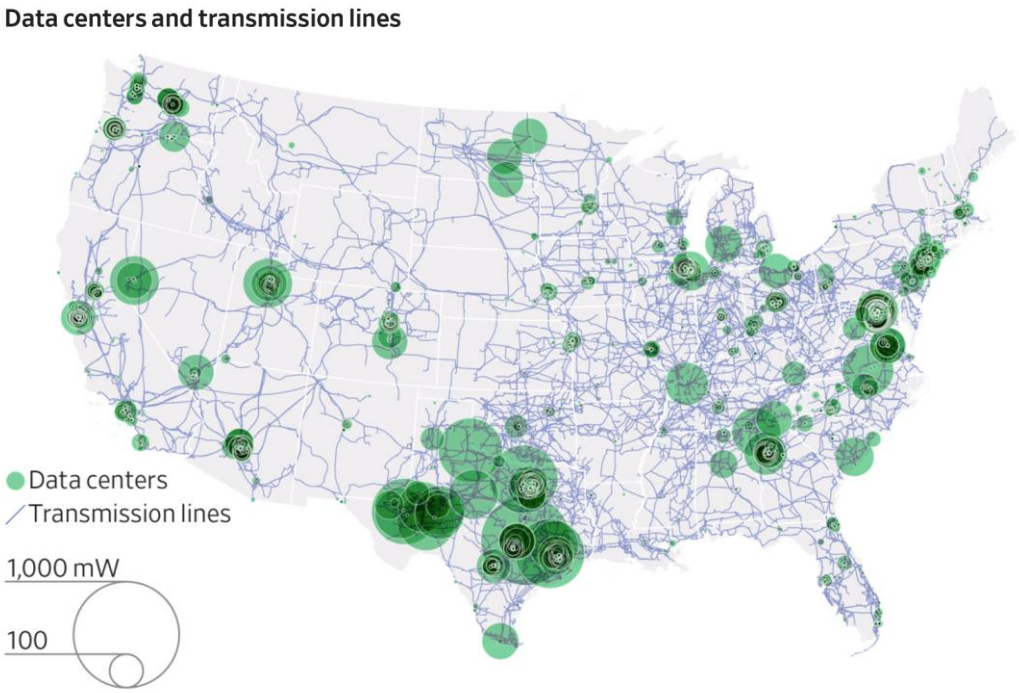
Real-Time Insights into Self-Learning Cycle



Onsite automation captures real-time insights
→ feeds the digital twin with predictive intelligence
→ which in turn refines and adapts the automation kit
→ creating a continuous self-learning cycle of the task

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AI-Driven Power Market Opportunity



AI Data Centers, Desperate for Electricity, Are Building Their Own Power Plants

Utilities are pushing to increase supply, but meeting the surge in demand won't be easy or fast. Tech companies aren't waiting around.

Supporting Points:

- Explosive AI growth is driving a surge in electricity demand that far exceeds grid capacity.
- Hyperscalers like OpenAI–Oracle ("Stargate") and xAI ("Colossus") are building on-site power plants to stay operational.
- "Bring Your Own Power" is emerging as the new model — faster, more reliable, and increasingly essential.

→ Implications for Xpanner:

- Rapid expansion of on-site energy and data infrastructure construction is accelerating demand for automation and hilgh-efficiency equipment.
- Xpanner's Physical A solutions are well-positioned to support these projects with scalable, autonomous field deployment capabilities.
- The trend validates Xpanner's focus on mission-critical, field-proven automation that bridges AI infrastructure and physical operations.

Source: WSJ (https://www.wsj.com/business/energy-oil/ai-data-centers-desperate-for-electricity-are-building-their-own-power-plants-291f5c81?st=M7cS76&reflink=article_copyURL_share)

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Forbes Highlights Xpanner's Role in Building the Future of Energy Infrastructure

Forbes Subscribe: Less than $1.50/wk

INNOVATION > TRANSPORTATION

Physical AI - Building The Energy Infrastructure For Data Centers

By Sabbir Rangwala, Senior Contributor. I write about... Follow Author

Published Oct 01, 2025, 11:00am EDT, Updated Oct 06, 2025, 12:51pm EDT

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Xpanner, "Redefining Construction with AI-Powered Automation"

Founded in 2020 in South Korea (the company name is motivated by the spanner, a common and critical construction tool) with $27M in investments, the company located to California in 2023, and is headed by CEO Henri Lee, who gained significant experience in construction at companies like Bobcat (acquired by Doosan Heavy Industries in 2008) and Hyundai Infracore. It brands itself as an ***"Automation as a Service"*** provider for the construction industry, is profitable, and to date has completed 23,000 autonomous hours serving customers across the globe. It has achieved revenue growth of ~250% over the past 2 years, primarily fueled by the growth of data center and energy project construction. It expects to book $21M in revenues across 14 projects in 2025 (currently at $17M). Currently, it has 65 employees based in South Korea and the United States.

Per Mr. Lee, Xpanner customers are looking to address skilled labor shortages, improve operational efficiency and safety, and very importantly, provide real-time project status information to customers (typically construction companies) and their customers (AI giants, energy businesses, land developers and government entities). Real time conversations and messages between on-site staff are also recorded and processed through LLMs for communicating problems and status. These capability are critical for ensuring that quality and build-to-print designs are being executed, gaps are corrected quickly, and unexpected issues escalated and resolved, minimizing expensive rework and schedule impacts.

Click the logo to read the full article

Forbes

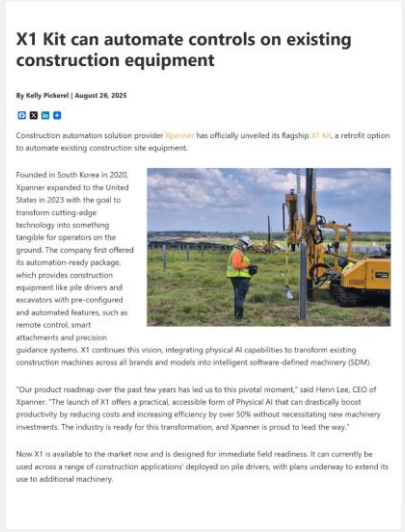
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Solar Power World

Leading U.S. publication covering solar installation, projects, and technology trends.



X1 Kit can automate controls on existing construction equipment

By Kelly Pickerel | August 26, 2025

Construction automation solution provider Xpanner has officially unveiled its flagship X1 Kit, a retrofit option to automate existing construction site equipment.

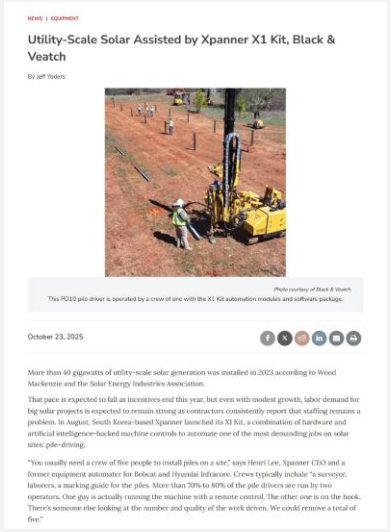
Founded in South Korea in 2020, Xpanner expanded to the United States in 2023 with the goal to transform cutting-edge technology into something tangible for operators on the ground. The company first offered its automation-ready package, which provides construction equipment like pile drivers and excavators with pre-configured and automated features, such as remote control, smart attachments and precision guidance systems. X1 continues this vision, integrating physical AI capabilities to transform existing construction machines across all brands and models into intelligent software-defined machinery (SDM).

"Our product roadmap over the past few years has led us to this pivotal moment," said Henri Lee, CEO of Xpanner. "The launch of X1 offers a practical, accessible form of Physical AI that can drastically boost productivity by reducing costs and increasing efficiency by over 50% without necessitating new machinery investments. The industry is ready for this transformation, and Xpanner is proud to lead the way."

Now X1 is available to the market now and is designed for immediate field readiness. It can currently be used across a range of construction applications' deployed on pile drivers, with plans underway to extend its use to additional machinery.

ENR Engineering News-Record

Leading U.S. construction and engineering media outlet covering infrastructure, technology, and innovations.



NEWS | EQUIPMENT

Utility-Scale Solar Assisted by Xpanner X1 Kit, Black & Veatch

By Jeff Yoders

Photo courtesy of Black & Veatch

This PD10 pile driver is operated by a crew of one with the X1 Kit automation modules and software package.

October 23, 2025

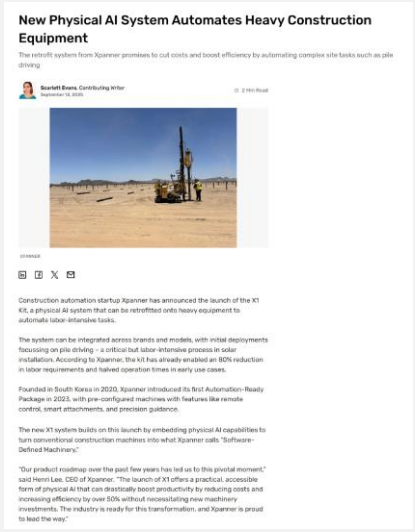
More than 40 gigawatts of utility-scale solar generation was installed in 2023 according to Wood Mackenzie and the Solar Energy Industries Association.

That pace is expected to fall as incentives end this year, but even with modest growth, labor demand for big solar projects is expected to remain strong as contractors consistently report that staffing remains a problem. In August, South Korea-based Xpanner launched its X1 Kit, a combination of hardware and artificial intelligence-backed machine controls to automate one of the most demanding jobs on solar sites: pile-driving.

"You usually need a crew of five people to install piles on a site," says Henri Lee, Xpanner CEO and a former equipment automator for Bobcat and Hyundai Infracore. Crews typically include "a surveyor, laborers, a marking guide for the piles. More than 70% to 80% of the pile drivers are run by two operators. One guy is actually running the machine with a remote control. The other one is on the hook. There's someone else looking at the number and quality of the work driven. We could remove a total of five."

AI⚙BUSINESS

Global media platform reporting on artificial intelligence innovations and enterprise adoption.



New Physical AI System Automates Heavy Construction Equipment

The retrofit system from Xpanner promises to cut costs and boost efficiency by automating complex site tasks such as pile driving.

Scarlett Evans, Contributing Writer

September 15, 2025

2 Min Read

STRANGER

Construction automation startup Xpanner has announced the launch of the X1 Kit, a physical AI system that can be retrofitted onto heavy equipment to automate labor-intensive tasks.

The system can be integrated across brands and models, with initial deployments focussing on pile driving - a critical but labor-intensive process in solar installation. According to Xpanner, the kit has already enabled an 80% reduction in labor requirements and halved operation times in early use cases.

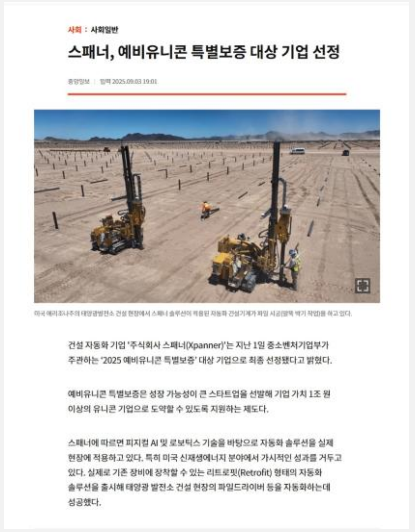
Founded in South Korea in 2020, Xpanner introduced its first Automation-Ready Package in 2023, with pre-configured machines with features like remote control, smart attachments, and precision guidance.

The new X1 system builds on this launch by embedding physical AI capabilities to turn conventional construction machines into what Xpanner calls "Software-Defined Machinery."

"Our product roadmap over the past few years has led us to this pivotal moment," said Henri Lee, CEO of Xpanner. "The launch of X1 offers a practical, accessible form of physical AI that can drastically boost productivity by reducing costs and increasing efficiency by over 50% without necessitating new machinery investments. The industry is ready for this transformation, and Xpanner is proud to lead the way."

The JoongAng

Major national newspaper with strong influence in Korean business and policy circles.



사회 : 사회일반

스패너, 예비유니콘 특별보증 대상 기업 선정

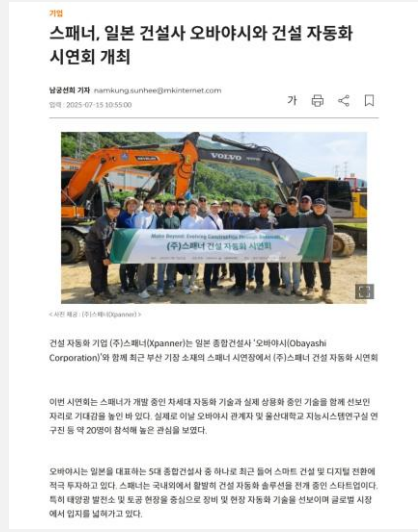
건설 자동화 기업 '주식회사 스패너(Xpanner)'는 지난 1일 중소벤처기업부가 주관하는 '2025 예비유니콘 특별보증' 대상 기업으로 최종 선정됐다고 밝혔다.

예비유니콘 특별보증은 성장 가능성이 큰 스타트업을 선발해 기업 가치 1조 원 이상의 유니콘 기업으로 도약할 수 있도록 지원하는 제도다.

스패너에 따르면 피지컬 AI 및 로보틱스 기술을 바탕으로 자동화 솔루션을 실제 현장에 적용하고 있다. 특히 미국 신재생에너지 분야에서 가시적인 성과를 거두고 있다. 실제로 기존 장비에 장착할 수 있는 리트로핏(Retrofit) 형태의 자동화 솔루션을 출시해 태양광 발전소 건설 현장의 파일드라이버 등을 자동화하는데 성공했다.

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기업

스패너, 일본 건설사 오바야시와 건설 자동화 시연회 개최

남궁선희 기자 namkung.sunhee@mkinternet.com

입력 : 2025-07-15 10:55:00

가

건설 자동화 기업 (주)스패너(Xpanner)는 일본 종합건설사 '오바야시(Obayashi Corporation)'와 함께 최근 부산 기장 소재의 스패너 시연장에서 (주)스패너 건설 자동화 시연회

이번 시연회는 스패너가 개발 중인 자세대 자동화 기술과 실제 상용화 중인 기술을 함께 선보인 자리로 기대감을 높인 바 있다. 실제로 이날 오바야시 관계자 및 울산대학교 지능시스템연구실 연구진 등 약 20명이 참석해 높은 관심을 보였다.

오바야시는 일본을 대표하는 5대 종합건설사 중 하나로 최근 들어 스마트 건설 및 디지털 전환에 적극 투자하고 있다. 스패너는 국내외에서 활발히 건설 자동화 솔루션을 전개 중인 스타트업이다. 특히 태양광 발전소 및 토공 현장을 중심으로 장비 및 현장 자동화 기술을 선보이며 글로벌 시장에서 입지를 넓혀가고 있다.

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Why now? – Macro Drivers

Theme	Key trends and implications	Opportunity
AI / Datacenter Surge	• Global datacenter power demand is expected to grow **175 % by 2030, equivalent to "adding another top-10 power-consumer country".** → US power demand CAGR = 2.6 % through 2030, with data centers making up 11 % of total electricity use.	***"These constraints make field automation a strategic necessity -- not an option"***
Labor Crisis	• Meeting this expansion requires ~510 K new US power & grid jobs and ~250 K in Europe, yet apprenticeship pipelines remain short by ~30 %. → **Transmission & Distribution (T&D) is the single largest bottleneck, demanding automation and productivity solutions.**	
Capex Wave	• **$790 B US grid investment + $40 B annual datacenter build through 2030** → structural demand for autonomous, high-precision field operations.	

Source: 1) Global Datacenter Supply/Demand update: Tight conditions likely to persist into 2026 + Read-across for Power, Hardware, and Industrial Tech / Engineering(13 October 2025)
2) Evaluating Labor Challenges for Generational Growth in US/Europe Power Demand (June 30, 2025)

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Target Verticals & Sequencing: High level

Xpanner's GTM aligns with the fastest-growing, most labor-strained construction verticals:
Target TAM: $1.0 T ("AI / Data / Energy Infrastructure Sectors").

Phase	Sector	Demand Catalyst	Market Tailwind
1	**Solar → Grid Transition**	• IRA-driven + grid connection boom (Before OBBBA) + Data Center	• Demand for automation has surged (piling, panel lifting, etc.) • Still increasing due to Data center demand
2	**BESS / Energy Storage**	• Grid balancing / micro-grid projects ↑ 260 B$ US transmission capex	• Urgent need for high-precision trenching & structure placement • Highly aligned with Solar
3	**Data Centers**	• AI-compute load + energy-intensive infrastructure	• $40 B annual construction spend (+20 % CAGR to 2027) • Most of Xpanner's existing clients are involved in data center construction, though at an early stage of adoption.
4	**Power Distribution & Grid Modernization**	• 750 K worker gap	• Structural demand for autonomous T&D equipment. • Automation demand mirrors the mining sector, offering an opportunity to extend into critical resource extraction.

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Target Verticals & Sequencing: Detail

Phase	Sector	Why This Order? (Strategic Rationale)
Solar / Renewable → Grid Transition	- **Proven use case:** Xpanner already achieved **57% cycle-time reduction** in PD10 pile-driving automation with top U.S. EPCs. - **Fastest ROI visibility:** Immediate labor replacement and precision benefits make solar automation the ideal pilot segment. - **Post-OBBBA transition:** Utility-scale solar slowed, while automation demand is **shifting toward hybrid BESS and grid-interconnection projects** . - **Existing customer base:** Ongoing deployments with Mortenson, Qcells, B&V, DEOPCOM, NoBull, etc..	- U.S. power demand CAGR **2.6%** through 2030; data centers = **11% of total U.S. electricity** .- **$260B** U.S. transmission investment through 2030 .
BESS / Energy Storage	- **Next logical step in the power network:** BESS is now the central node of renewable-to-grid transition; enabling both grid balancing and data-center backup power. - **High automation potential:** Trenching, piling, and cable container handling align with Xpanner's proven automation modules. - **Severe labor shortage:** T&D and BESS construction share the same skill bottleneck; automation mitigates up to **30% of labor gap, plus existing customer base.**	- U.S. grid workforce gap: **510K** by 2030 .- Transmission & distribution labor shortfall: **78K** skilled workers .
Data Center Construction	- **Explosive power demand:** Global data center electricity use up **175% by 2030**, equal to adding another top-10 power-consuming country . - **Timing advantage:** Supply remains tight until at least 2026 — driving strong demand for automated, precision field operations. - **Technology reusability:** Panel lifting, material handling, and grading modules from solar projects directly transfer to data center build; and **strong alignment with existing customer base.**	- Data center construction ≈ **$40B annually**, **20% CAGR** through 2027 .- Data centers = **11% of U.S. total power use** .
Power Distribution & Grid Modernization	- **Final stage of Physical AI adoption:** Extends automation from isolated machines to **networked, self-operating field systems** across the grid. - **Severe labor dependence:** Transmission and distribution are the most labor-intensive and time-sensitive infrastructure segments — automation becomes structural, not optional . - **AI OS integration:** Multiple sensors and equipment types unified under Xpanner's Physical AI OS.	- Transmission CapEx ≈ **$258B** through 2030 .- Grid automation market growing **>10% YoY**.

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Why This Sequence Makes Strategic Sense

Criteria	Principles
Technical & ROI Progression	• Start where automation is easiest to prove ROI (solar) → move to more complex, higher-value environments (BESS, data centers, grids). • **Each phase increases automation scope and data value.**
Aligned with CapEx Flows	• Investment naturally flows from Renewables → Energy Storage → Data Infrastructure → Power Grid after the OBBBA policy shift. • **Xpanner's roadmap follows this flow, not fights it.**
AI Learning Flywheel	• **Every phase adds operational data that enhances AI models;** enabling faster deployment, higher accuracy, and lower integration cost for the next vertical.

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Marketing and sales strategy: Lead-to-Revenue Strategy

Lead generation plan

1. Event-Driven Pipeline Growth

- Industry events such as **RE+** and targeted **regional showcases** generated the highest-quality leads, directly contributing to 10**+ sales opportunities**
- On-site demos significantly improved conversion from interest → pilot deployment

2. Client-Driven Virality

- Referrals and visibility from existing EPC clients remain the most effective and cost-efficient source of high-quality enterprise leads, with the highest conversion rate across all channels.

3. PR & Thought Leadership

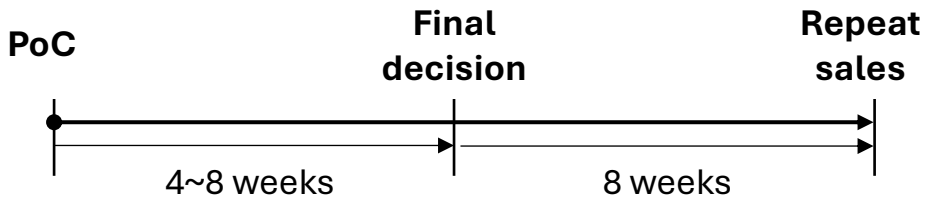
- Strategic media coverage and case-study publications generated **50+** qualified inbound leads, reinforcing credibility in Physical AI and automation domains.

4. Digital Marketing Acceleration

- Optimized content and audience targeting expanded inbound pipeline coverage.
- Performance marketing efficiency improvement

Turning Interest into Impact

- ***PoC is the fastest and most credible conversion engine, and every successful demo becomes a new long-term revenue stream.***



1. PoC-Centric Sales Motion

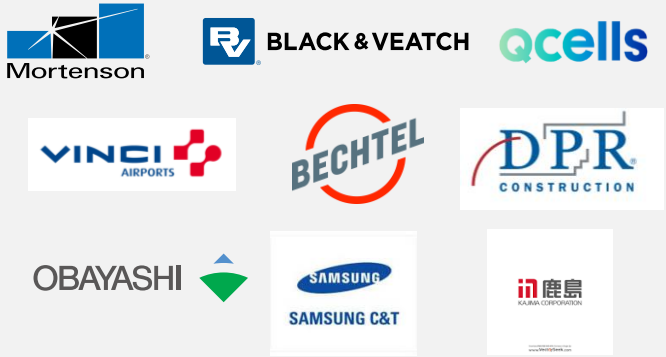
- PoC deployments serve as the most persuasive stage — allowing clients to directly experience Xpanner's automation ROI (cycle-time, accuracy, cost savings).
- The average time from PoC to purchase decision is significantly shorter than industry benchmarks, creating a faster revenue conversion cycle.

2. Rapid Repurchase Cycle

- Once validated, EPCs and developers immediately repeat orders for subsequent projects, turning one PoC into a multi-project rollout.
- This model builds high retention and compound account growth with minimal additional CAC.

Channel & Partnership Play

EPC Alliances



Recurrent deployment base.

OEM Integration



Standardized API interface for retrofit.

Tech Ecosystem



Digital Twin integration.

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Data Loop & Automation Flywheel

- **Each X1/X2 deployment collects multi-modal GNSS + IMU + video data, feeding Xpanner's AI models.**



Regional Execution Roadmap

Stage	Region	Focus
2025–26	U.S.	Renewables + Datacenter construction (80 % revenue base)
2026–27	Japan / Korea	BESS / grid modernization projects
2027–28	Australia / Europe	Data + Energy infrastructure automation scale-up

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Key milestones

ARR >$50 M
(>70 % recurring)

1,500 + machines retrofitted

10 + Physical AI modules commercialized

IPO ready by 2028 on NASDAQ —

Positioned as Physical AI standard in construction & energy infrastructure

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Unit economics

Revenue Models Overview

Model	Description	Pricing	Customer side impact
Perpetual (CapEx)	One-time sale of Physical AI Module (HW + lifetime SW license), but new HW will release in 2~3 years causing new license purchase.	**$90K / unit**	-Corporate budget(CapEx) -Longer time to get an approval
Subscription (AaaS)	Monthly service including HW lease, SW, OTA, field ops & support.	**$7K / month**	-Project budget(Expenses) -Quick decision by a site manager

Unit Economics Comparison

Metric	Perpetual Sale	Subscription (AaaS)
Unit Price	$90,000 (one-time)	$7,000 / month
Hardware COGS	$14,000	$14,000 (amortized)
Gross Margin	**84%**	**94%**
Gross Profit	**$76K (1 Year)**	**$238K (3 years)**
Customer Retention	One-time	36 months (avg)
CAC (incl. PoC)	$10K	$10K
LTV / CAC**	7.6x	23.8x
Payback Period	**Immediate**	**< 2 months**

*CAC: 10K per customer (including all costs for F2F PoC)
**LTV: Gross Profit